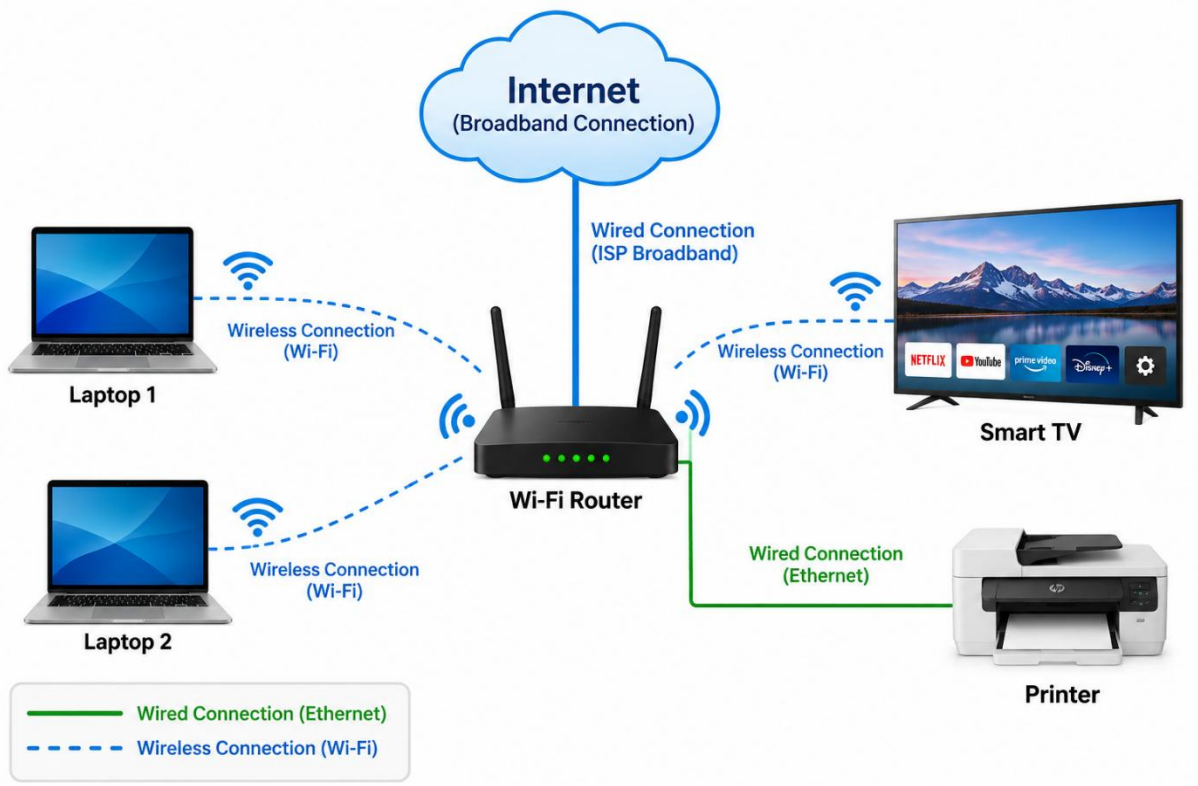
1) Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless)



**2)List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

1. Recommended hardware and models

These are practical model suggestions for the Indian market, with examples from current online listings. Prices and availability can change.



1. Wi-Fi router — 1 unit

TP-Link Archer C6 AC1200

Purpose: Shares the broadband connection with all devices.

Features: 2.4 GHz and 5 GHz Wi-Fi, Ethernet WAN and LAN ports.

Approximate budget: ₹2,500–₹3,500.

2. Network switch — optional, 1 unit



TP-Link TL-SG105

- Purpose: Adds Ethernet ports for more wired devices.
- Not required for the basic setup if the router has a free LAN port for the p
- Approximate budget: ₹1,200–₹2,500.



3. Laptops — 2 units

Suggested: HP 15s series or Lenovo IdeaPad Slim 3.

Purpose: Office work, browsing, email and file sharing.

Connectivity: Wireless Wi-Fi.

Check that each laptop has a compatible Wi-Fi adapter; most modern laptops do.

5. Printer — 1 unit



HP Smart Tank 580

- Purpose: Printing documents from either laptop.
- Important: This model supports Wi-Fi, so it can also connect wirelessly. For the diagram, use Ethernet only if the chosen printer has a compatible Ethernet port.
- If you specifically require a wired network printer, choose a model with an RJ45 LAN port, such as a suitable HP OfficeJet Pro or business laser printer.

2) Cables and supporting accessories

Item	Quantity	Recommended specification	Purpose
Broadband fibre/ONT connection	**1**	**ISP-supplied ONT or modem, if required**	**Connects the office to the internet**
Cat6 Ethernet patch cable	**1**	**Length based on router-to-printer distance; typically 2–5 m**	**Wired printer connection**
Cat6 Ethernet cable	**1**	**Length based on ISP device-to-router distance**	**Connects ONT/modem to router WAN port, if separate**

Router power adapter	**1**	**Manufacturer-supplied, correct voltage**	**Powers the router**
Printer power cable	**1**	**Compatible with the printer**	**Powers the printer**
Surge-protected power strip	**1**	**BIS-compliant, suitable electrical rating**	**Powers and protects networking equipment**
Cable clips or ties	**1 pack**	**Suitable for cable routing**	**Keeps cables organised**

3) Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

step 1: Connect to your Airtel Wi-Fi router

1. Switch on your Airtel Xstream Fiber router.
2. Connect your laptop or mobile phone to the existing Airtel Wi-Fi network, or connect the laptop using a LAN cable.
3. Keep the router powered on during configuration.

Step 2: Log in to the Airtel router

Common Airtel router gateway

192.168.1.1

Step 3: Change the Wi-Fi network name (SSID)

1. Find Setup → WLAN → Basic, Wireless Settings, or a similar menu.
2. Locate the SSID or Wi-Fi Name field.
3. Enter a unique network name, for example:
    - OfficeNet-24 for 2.4 GHz
    - OfficeNet-5G for 5 GHz
4. If your router combines both bands, set one name such as OfficeNet.
5. Click Apply or Save.

Step 4: Enable WPA2 security

1. **Open WLAN → Security, Wireless Security, or the equivalent menu.**
2. **Select the following settings if available:**

Setting	Value
Authentication / Security	WPA2-PSK or WPA2-Personal
Encryption	AES
Wi-Fi password	A unique, strong password
WPS	Disabled unless required

3. **Enter your new Wi-Fi password.**
4. **Click Apply or Save.**

If the router supports WPA3 and all your devices are compatible, WPA3-Personal is preferable. Otherwise, WPA2-AES is a good compatibility choice. Do not select WEP or TKIP.

Step 5: Change the default administrator password

1. Open Maintenance → Password, System → Administration, or a similar menu.
2. Select the administrator account and choose Modify, if shown.
3. Enter the current administrator password.
4. Enter a new, strong administrator password and confirm it.
5. Save the changes and log in again with the new password.

Use a different password from your Wi-Fi password. If the administrator password option is missing or locked by Airtel, contact Airtel support rather than resetting the router unnecessarily.

Step 6: Reconnect your SOHO devices

- Laptop 1: Connect to the new SSID and enter the Wi-Fi password.
- Laptop 2: Connect to the same network.
- Smart TV: Open its network settings and reconnect to Wi-Fi.
- Printer: Connect via Ethernet to a router LAN port, provided the printer has an Ethernet interface.

4) Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device IP address allocation table

The MAC addresses below are fictional examples for an assignment; replace them with the real addresses from each device.

Device	MAC address (example)	Assigned IP address	Method
Airtel Wi-Fi router	02:11:22:33:44:01	192.168.1.1	Static / gateway
Laptop 1	02:11:22:33:44:02	192.168.1.100	DHCP
Laptop 2	02:11:22:33:44:03	192.168.1.101	DHCP
Smart TV	02:11:22:33:44:04	192.168.1.20	Static
Printer	02:11:22:33:44:05	192.168.1.10	Static

5) Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible causes and solutions

No.	Possible cause	Diagnosis and solution
1	Wi-Fi is disabled or incorrect password	Check that Wi-Fi is enabled and Airplane Mode is off. Confirm that the laptop is connected to the correct SSID and enter the correct Wi-Fi password.
2	Incorrect IP address or DHCP failure	Open Command Prompt and run ipconfig. If the laptop has an address beginning with 169.254, it may not have received an IP address from the router. Run ipconfig /release followed by ipconfig /renew.
3	Airtel router or internet connection problem	Check the router's power and internet indicators. Test the internet on another device. If all devices fail, restart the router and ONT if applicable. Contact Airtel support if the connection remains down.
4	Weak Wi-Fi signal or interference	Move the laptop closer to the router. Try the 2.4 GHz Wi-Fi band if the 5 GHz signal is weak, and reconnect to the network.
5	DNS configuration problem	If the laptop connects to Wi-Fi but websites do not load, run nslookup google.com. If name resolution fails, check the DNS settings and set them to automatic or a trusted DNS server.